

Advantages of Lithium-air Batteries

Characteristics	Lithium-air battery
Capacity	Ten times more capacity than Lithium-ion batteries
Weight	Lighter than Li-ion and other cells, as the cathode is oxygen
Cost	Less, because of use of Li and cathode of it is oxygen which is absorbed from air
Effect on environment	Environment-friendly

current than Li-air can provide (it is a high-energy design), so a hybrid system using Li-air and another energy device linked electronically will be used. Hybrid Li-air systems will be the next battery after lithium-air, or these may be developed simultaneously in order to leverage the significant energy density advantages already demonstrated.

James G. Bellingham, PhD, chief technologist, Monterey Bay Aquarium Research Institute, says, "The lithium-seawater battery can provide much more energy for the same weight than Lithium-ion. The lithium-air battery uses similar technology to the lithium-seawater, but the lithium-air battery is designed for air environment, and is not appropriate for being submerged

Also, there are developments happening other than the Lithium-air battery. PolyPlus Battery, a privately held company located in California, is working on Lithium-seawater batteries in conjunction with Monterey Bay Aquarium Research Institute.

in the ocean."

Bellingham adds, "We are increasingly dependent on robots and autonomous instruments for exploring the undersea environment. Our big problem today is that the vehicles can only last a short time and cannot carry many sensors. The great promise of these batteries is that we will be able to make systems that last weeks or even months in the ocean, greatly expanding our ability to study and understand the ocean. Of course, many other human endeavours in the ocean depend on power as well, including such large enterprises as defence and the oil and gas industry; so expect the impact of such a greatly improved source of power." ●

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